John T. Fields

Business Analytics · Applied AI · Higher-Education Data Science

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Profile

Business analytics educator and applied-AI researcher who founded Concordia University Wisconsin's undergraduate Business Analytics degree program and co-founded its AI & Quantum Innovation Lab. Background combines a 29-year career at Rockwell Automation (1989–2018), spanning sales engineering through director-level leadership in enterprise data and sales operations, with peer-reviewed research in privacy-preserving machine learning, large language models, and student-success analytics. Teaches across business and computing (analytics, MIS, information systems, and computer science) and builds programs from the ground up, with an ethics- and faith-informed approach to technology.

Education

Ph.D., Computer Science, Marquette University Expected Dec 2026

Milwaukee, Wisconsin · Areas of emphasis: artificial intelligence, NLP, privacy-preserving machine learning

- Dissertation (in progress): privacy-preserving machine learning for cross-institutional student-retention prediction. Advisor: Dr. Praveen Madiraju. Defense scheduled September 2026; degree conferred December 2026.

M.S., Applied Data Science, Syracuse University 2020

GPA 3.9

B.S., Industrial Distribution, Texas A&M University 1989

College of Engineering

Academic Appointments

Assistant Professor of Business Analytics 2020 – Present

Concordia University Wisconsin · Batterman School of Business

- Founded Concordia's undergraduate Business Analytics degree program—developed all courses, the course sequence, and the integrative capstone; this curriculum later served as the foundation for the M.S. in Business Analytics.
- Teach undergraduate and graduate courses across business and computing: business analytics programming and visualization, analytics and the digital economy, business statistics, and management information systems.
- Mentor undergraduate and graduate research, including projects in NLP and privacy-preserving machine learning.

Co-Founder & Chair, AI & Quantum Innovation Lab 2025 – Present

Concordia University Wisconsin

- Co-founded an interdisciplinary lab advancing applied AI and quantum computing within a faith-informed academic setting.
- Direct student and faculty projects spanning privacy-preserving ML, NLP, and edge AI; organize the lab's research symposia and public outreach.

Professional Experience (Industry)

Data Scientist, Talent Select AI (formerly HarQen) 2019 – 2023

Milwaukee, WI / New York, NY

- Built and deployed NLP and machine-learning models for talent assessment and selection, used by enterprise clients.
- Contributed to product development of AI-driven hiring and interview-analysis tools.

Rockwell Automation 1989 – 2018

Milwaukee, Wisconsin

- Director, Global Customer Data (2013–2018): led enterprise customer master-data strategy and information governance across global operations; received the SAP IGgie Award (2015).
- Director, Market-to-Quote Processes (2008–2012): directed go-to-market process design and integration across sales operations.
- Manager, Channel Operations (2004–2007): managed North American distribution-channel operations and partner programs.
- Sales engineer roles (1989–1999), then leadership roles in IT and sales operations (1999–2004).

Research Interests

Privacy-preserving machine learning · large language models and NLP · educational data mining and student-success analytics · data literacy · AI ethics and algorithmic fairness.

Research Metrics

Google Scholar (June 2026): ~228 citations. Lead author of a highly cited IEEE Access survey on transformer-based text classification.

Publications

Refereed Journal Articles

Fields, J., Chovanec, K., & Madiraju, P. (2024). A survey of text classification with transformers: How wide? How large? How long? How accurate? How expensive? How safe? *IEEE Access, 12,* 6518–6531. https://doi.org/10.1109/ACCESS.2024.3349952

Refereed Conference Proceedings

Chovanec, K., **Fields, J.**, & Madiraju, P. (2025). Explaining pre-trained language models in the context of higher education. *IEEE International Conference on Big Data (IEEE BigData 2025).*

Islam, K. M. S., **Fields, J.**, & Madiraju, P. (2025). MultiMentalRoBERTa: A fine-tuned multiclass classifier for mental-health disorder. *IEEE International Conference on Big Data (IEEE BigData 2025).* (arXiv:2511.04698)

Fields, J., Chovanec, K., & Madiraju, P. (2024). Integrating categorical and continuous data in a cluster-then-classify methodology for predicting undergraduate student success. *IEEE International Conference on Big Data (IEEE BigData 2024),* 8118–8126. https://doi.org/10.1109/BigData62323.2024.10825671

Chovanec, K., **Fields, J.**, & Madiraju, P. (2023). Combining demographic tabular data with BERT outputs for multilabel text classification in higher-education survey data. *IEEE International Conference on Big Data (IEEE BigData 2023),* 1403–1409. https://doi.org/10.1109/BigData59044.2023.10386843

Under Review & In Preparation

Fields, J., Islam, K. M. S., Thota, R., Chen, V., & Madiraju, P. (2025). A privacy-preserving framework for cross-institutional collaboration on student-retention prediction using remote data science. *Under review, IEEE International Conference on Information Reuse and Integration (IEEE IRI).*

Clark, J., Bennett, E. E., & **Fields, J.** How artificial intelligence will transform project management and the work of HRD. In A. Alizadeh, K. Dirani, & J. Li (Eds.), *Artificial Intelligence and HRD.* Palgrave Macmillan. (Book chapter, under review.)

Grants & Sponsored Research

Fields, J. (PI). Privacy-Preserving Machine Learning for Improving University Student Retention. National Science Foundation, NAIRR Pilot (Award NAIRR240195), 2024–2025.

Patents

Fields, J. Machine-learning and text-mining methodology for higher-education admissions decisioning. U.S. Provisional Patent Application 62/935,928.

Invited Talks, Panels & Presentations

Fields, J. Privacy-preserving machine learning in education. Georgetown Massive Data Institute Summer Institute, Washington, DC, 2025.

Fields, J., & Thota, R. Privacy-preserving machine learning for student retention. NAIRR Annual Meeting, Washington, DC, 2025.

Fields, J. Predicting student dropout in higher education: A cluster-label-classify approach. Wisconsin Interdisciplinary Research Symposium, Mequon, WI, 2025.

Fields, J. Integrating categorical and continuous data in a cluster-then-classify methodology for predicting undergraduate student success. IEEE International Conference on Big Data, Washington, DC, 2024.

Fields, J. (panelist), with Dr. Robert J. Marks II (Baylor University). Panel discussion on AI. Inspire & Ignite Leadership Conference, Mackinac Island, MI, 2024.

Fields, J. Privacy-preserving machine learning in education. Accelerating the Equitable Use of Education Data Forum, Georgetown University (online), 2024.

Fields, J. The future of text classification beyond ChatGPT. Inspire & Ignite Leadership Conference, Mackinac Island, MI, 2024.

Fields, J. Combining demographic tabular data with BERT outputs for multilabel text classification. IEEE International Conference on Big Data, Sorrento, Italy, 2023.

Teaching

Courses taught at Concordia University Wisconsin (undergraduate and graduate):

- Business Analytics Programming with Visualization (BUAN 4850 / BUS 405)
- Analytics and the Digital Economy (BUAN 4900 / BUS 410)
- Data Analytics: Integrative Project, Capstone (BUAN 4950 / BUS 415)
- Business Statistics (BUS 315)
- Management Information Systems (BUS 355 / BUS 3420), 20+ sections
- Principles of Management (MGMT 130)

Research Mentorship

- Supervise undergraduate and graduate research through the AI & Quantum Innovation Lab.

Awards & Honors

- Undergraduate Business Faculty of the Year, Batterman School of Business, Concordia University Wisconsin (2026)

- Faculty Award for Faith and Learning, Concordia University Wisconsin (2024)
- Undergraduate Business Faculty of the Year, Batterman School of Business, Concordia University Wisconsin (2024)
- SAP IGgie Award for Information Governance, Rockwell Automation (2015)

SERVICE

University & College

- Faculty AI Committee (2026–present)
- Search committees: Batterman School of Business Dean (2024–2025); Computer Science (2025–2026) and Finance/Accounting (2026) faculty hires
- Faculty Student Retention Committee (2023–2024); Predictive Analysis Committee (2021–2023); Faculty Advisor, Strategic Planning (2023–2024)
- Faculty representative at CUW admissions and campus-visit events (2021–2025)

Professional

- Chair & Program Coordinator, AI & Quantum Innovation Lab (2025–present); reviewer, IEEE IRI (2025)

Church & Community

- Sunday-school / Bible-study teacher, Life Church Milwaukee (2023–present); food-pantry task force, Life Center Milwaukee (2020–present)
- Guest speaker at Lutheran high schools in Wisconsin (2020–2021)

PROFESSIONAL MEMBERSHIPS & CERTIFICATIONS

- IEEE (2023–present); ACM (2021–present); the Veritas Forum (2020–present); the Cambridge Roundtable on Science and Religion (2020–2021)
- CITI Social & Behavioral Research, Investigator Certification (2024–2028); DataRobot Automated Machine Learning I (2020)

PUBLIC SCHOLARSHIP & OUTREACH

- Coordinated monthly AI & Quantum Innovation Lab events during the academic year, including guest speakers on quantum computing and AI ethics/philosophy, community engagement, and a U.S. State Department–coordinated visit by international journalists (2025–2026); continuing-education AI video/webinar series (2025–2026).